

DEEP LEARNING · STRUCTURAL MRI · ALZHEIMER'S DISEASE

Neurological Condition Classification from Structural MRI

Deep Learning Approaches for CN, MCI, and AD Detection



Task 3: Multi-class (CN/MCI/AD)

Task 2: Binary Classification

- Problem Setup & Motivation
- Slice-based MRI Representation
- Preprocessing Pipeline
- ResNet-18 Architecture
- Training & Cross-Validation
- Ensemble & Evaluation
- Grad-CAM Interpretability

Task 3: Multi-class Classification

- Problem Setup & Challenges
- Data Representation
- SimCLR Self-supervised Pretraining
- Contrastive Loss Formulation
- Supervised Fine-Tuning
- Evaluation & Results
- Clinical Significance

TASK 2 · BINARY CLASSIFICATION

CN vs AD Detection

Task 2 — Problem Setup & Motivation

Classification Goal

Distinguish **Cognitively Normal (CN)** subjects from **Alzheimer's Disease (AD)** patients using structural MRI.

Key Challenges:

- 3D MRI volumes are computationally expensive
- High inter-subject variability
- Class imbalance in clinical datasets
- Need for interpretable predictions

Our Approach:

Slice-based 2D deep learning with transfer learning,

CN

Cognitively Normal

Binary

AD

Alzheimer's Disease



3D MRI Input

Task 2 — Slice-based MRI Representation

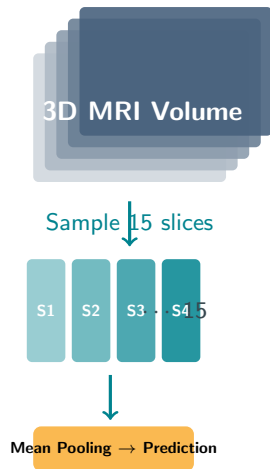
Why 2D Slices?

Processing full 3D volumes is computationally intensive.

Slice-based approach balances efficiency & spatial coverage.

Extraction Strategy:

- **15 slices** extracted per volume
- **Uniform sampling** along the depth axis
- Ensures consistent spatial coverage across all subjects
- Each slice processed independently through backbone
- Slice predictions aggregated via **mean pooling**



Task 2 — Preprocessing Pipeline

Standardization Steps:



Intensity Clipping

Clip to [1st, 99th] percentile to remove outliers



Z-score Normalization

Normalize to zero mean and unit variance

$$x' = \frac{x - \mu}{\sigma}$$

Min-Max Scaling (per slice)

Scale each slice to [0, 255] for model compatibility



Channel Expansion

Grayscale → 3-channel for pretrained model input

Data Augmentation (Train Only)

- ✓ Random horizontal flip
- ✓ Small-angle rotation ($\pm 10^\circ$)

Validation / Test

No augmentation applied
Ensures unbiased evaluation

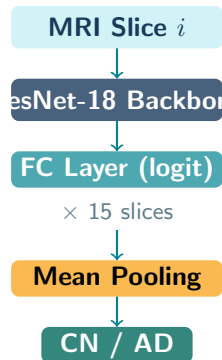
Key Benefit

Reduces inter-scan variability and ensures compatibility with ImageNet-pretrained backbone

Task 2 — Model Architecture (ResNet-18)

Backbone: ResNet-18 (ImageNet pretrained)

- Final FC layer replaced with **single logit output**
- Each of 15 slices processed **independently**
- Slice-level logits aggregated via **mean pooling**
- Subject-level prediction via sigmoid activation



Design Advantages:

- Captures **local structural patterns** per slice
- Aggregates **global context** across brain volume

Task 2 — Training Strategy

Cross-Validation

- **5-fold Stratified K-Fold**
- Class distribution preserved across folds
- Reduces overfitting and variance
- Results in **5 trained models** for ensemble

Early Stopping

- Monitored by **validation AUC**
- Triggered after **5 epochs** without improvement
- Best model (highest val AUC) saved per fold

Ensemble Inference

Predictions from all **5 fold models** are averaged:

$$\hat{y} = \frac{1}{5} \sum_{k=1}^5 \hat{y}_k$$

Benefits:

- Reduces prediction variance
- Enhances generalization
- Stabilizes outputs across test samples

Threshold Optimization

Search over $[0.1, 0.9]$ to maximize **balanced accuracy** (vs fixed 0.5)

Task 2 — Evaluation Metrics & Results

Evaluation Metrics:

AUC	Area Under ROC Curve
Bal. Acc	Balanced Accuracy
F1	Macro F1-score
P / R	Per-class Precision & Recall
CM	Confusion Matrix

Key Observations

- Ensemble learning stabilizes predictions
- Threshold tuning improves balanced accuracy
- Strong discriminative capability from structural MRI alone

Result Summary:

AUC 0.889

Balanced Acc 0.868

Macro F1 0.868

Precision [0.889, 0.846]

Threshold Optimization

Optimal threshold selected by maximizing balanced accuracy over [0.1, 0.9]

Task 2 — Interpretability via Grad-CAM

What is Grad-CAM?

Gradient-weighted Class Activation Mapping

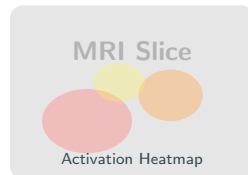
Highlights spatial regions contributing to model predictions.

Implementation:

- Applied to the **final convolutional layer**
- Gradients of predicted class w.r.t. feature maps
- Produces heatmap overlaid on MRI slice

Key Insights:

- ✓ Focuses on **anatomically meaningful regions**
- ✓ Avoids noise artifacts
- ✓ Provides visual evidence for decisions



• High activation

• Medium

• Low activation

relevant regions identified

Task 2 — Strengths, Limitations & Clinical Relevance

Strengths

- Efficient 2D approach avoids heavy 3D computation
- Transfer learning improves convergence
- Cross-validation + ensembling reduce overfitting
- Grad-CAM adds transparency
- Clinically deployable

Limitations

- May lose inter-slice 3D spatial relationships
- Mean aggregation assumes equal slice importance
- Slice selection strategy can impact performance

Clinical Relevance

- Structural MRI alone is discriminative for AD
- Lightweight model supports real-world deployment
- Aligns with early, scalable neurological screening

★ Binary classification demonstrates that structural MRI contains sufficient information for acc

TASK 3 · MULTI-CLASS CLASSIFICATION

CN vs MCI vs AD Detection

Task 3 — Problem Setup & Challenges

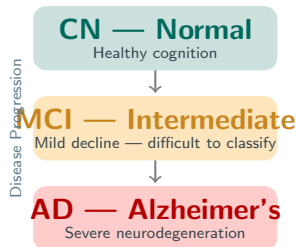
Three-Class Classification Goal

Classify subjects into:

- **CN** — Cognitively Normal
- **MCI** — Mild Cognitive Impairment
- **AD** — Alzheimer's Disease

Key Challenges:

- MCI is an **intermediate** and ambiguous state
- Subtle inter-class differences in MRI structure
- Class imbalance across CN, MCI, and AD
- Standard supervised learning may not capture fine-grained features



Task 3 — Data Processing & Representation

Depth Standardization

MRI volumes standardized to **fixed depth of 128 slices**:

- **Central cropping** when depth > 128
- **Symmetric zero-padding** when depth < 128

Slice Extraction

16 contiguous central slices extracted

Focuses on anatomically significant regions associated with neurodegeneration

Augmentation Strategy (Training)

- Random spatial flips (horizontal & vertical)

Raw MRI Volume (var. depth)

Standardize to 128 slices

Extract 16 Central Slices

Augment (2 views for SSL)

Ready for SimCLR / Fine-tuning

Task 3 — Self-Supervised Pretraining (SimCLR)

Why Self-Supervised Learning?

- Learn rich MRI representations **without labels**
- Reduces dependence on annotated data
- Improves convergence in supervised fine-tuning

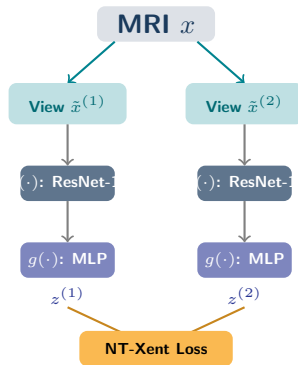
SimCLR Framework:

Given MRI volume x , generate two augmented views:

$$\tilde{x}^{(1)}, \tilde{x}^{(2)}$$

Passed through encoder $f(\cdot)$ and projection head $g(\cdot)$:

$$z^{(k)} = g\left(f\left(\tilde{x}^{(k)}\right)\right), \quad k \in \{1, 2\}$$



Task 3 — Contrastive Loss Formulation

Step 1: Normalize embeddings

$$\hat{z} = \frac{z}{\|z\|}$$

Step 2: Cosine similarity

$$\text{sim}(z_i, z_j) = \frac{z_i \cdot z_j}{\|z_i\| \|z_j\|}$$

Step 3: NT-Xent loss for positive pair (i, j)

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}$$

Step 4: Total loss over batch of size N

Temperature τ

Controls the sharpness of similarity distribution:

- Low τ : sharper, harder negatives
- High τ : smoother, easier negatives

Objective

Pull together representations of the **same volume** (different augmentations) while pushing apart **different volumes** in the embedding space.

Task 3 — Supervised Fine-Tuning

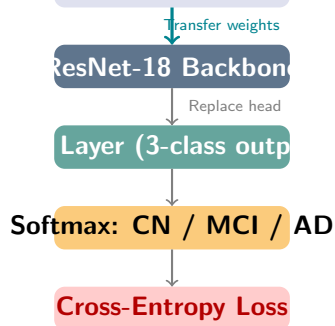
Knowledge Transfer

- 1 **Projection head removed** from pretrained encoder
- 2 **FC layer added** for 3-class prediction (CN / MCI / AD)
- 3 Backbone weights initialized from SimCLR pretraining

Training Configuration:

Loss Function	Cross-Entropy Loss
Optimizer	AdamW
Learning Rate	1×10^{-4}
Cross-Validation	5-fold Stratified K-Fold

Stage 1: SimCLR Pretraining



Task 3 — Evaluation Metrics & Results

Primary Metrics:

- **Multi-class AUC** (One-vs-Rest strategy)
- **Balanced Accuracy**

Why these metrics?

- Handle class imbalance across CN, MCI, AD
- Ensure fair evaluation of intermediate class (MCI)

Key Observations

- Self-supervised pretraining improves convergence & generalization
- CN and AD effectively distinguished
- **MCI remains challenging** — reflects real clinical ambiguity

Result Summary:

Metric	Score	Notes
AUC (OvR)	0.957	Strong separability
Bal. Acc	0.803	Handles imbalance well
CN F1	0.968	Very high performance
MCI F1	0.667	Challenging class
AD F1	0.810	Good detection

(Best fold: 4 Epoch: 36 Macro F1: 0.815)

MCI Insight

MCI classification difficulty mirrors genuine **clinical ambiguity** — consistent with literature.

Task 3 — Discussion & Clinical Significance

↑ Improvements over Task 2

- Contrastive self-supervised pretraining
- Rich, invariant MRI representations
- Better handling of subtle class boundaries
- Less dependence on labeled data

⚠ Limitations

- Slice-based: may miss full 3D spatial dependencies
- Basic augmentations limit contrastive potential
- MCI inherently difficult (overlapping features)

🏥 Clinical Significance

- SSL enhances medical imaging models
- Structural MRI models disease progression stages
- Supports early neurological screening
- Scalable and accessible pipeline

★ Self-supervised learning significantly enhances multi-class neurological condition classification

Pipeline Comparison: Task 2 vs Task 3

Component	Task 2 (Binary)	Task 3 (Multi-class)
Classes	CN vs AD	CN vs MCI vs AD
Backbone	ResNet-18	ResNet-18
Pretraining	ImageNet (supervised)	SimCLR (self-supervised)
Slices	15 (uniform)	16 (central)
Aggregation	Mean pooling	Mean pooling
Loss	BCE with Logits	Cross-Entropy
Cross-Val	5-fold stratified	5-fold stratified
Ensemble	5-model average	Per-fold models

CONCLUSIONS

Key Takeaways

Task 2 (CN vs AD)

Lightweight slice-based framework with transfer learning, ensembling, and Grad-CAM achieves strong binary classification from structural MRI alone.

Task 3 (CN/MCI/AD)

SimCLR-based self-supervised pretraining enables richer feature learning, improving multi-class differentiation especially for subtle MCI cases.

Both pipelines demonstrate the feasibility of scalable, accessible, MRI-based neurological screening systems.